

# Sadeep Ariyarathna

## Systems Engineer | Mechatronics Engineer

### Personal Information

|           |                                               |
|-----------|-----------------------------------------------|
| Address   | Wiedenstraße 7, Steinhagen, 33803 Germany     |
| Telephone | +49 1521 3746490                              |
| E-mail    | sadeepari@gmail.com                           |
| LinkedIn  | <a href="#">Sadeep Ariyarathna   LinkedIn</a> |
| Website   | <a href="#">Sadeep Ariyarathna</a>            |

### Experience

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2024 – 08/2025 | <p>CLAAS E-Systems, Dissen am Teutoburger Wald, Germany</p> <p><b>Intern– E/E Architecture</b></p> <ul style="list-style-type: none"><li>- Modelling system architecture in Enterprise Architect, focusing on system engineering concepts.</li><li>- Writing master thesis on collaboration techniques for merging and reviewing architecture models.</li></ul>                                                                                                                                                                                   |
| 09/2022 – 02/2023 | <p>AtlasLabs (Pvt.) Ltd, Sri Lanka</p> <p><b>Mechatronics Engineer</b></p> <ul style="list-style-type: none"><li>- Engineered and tested mechanical components and systems from concept to launch with Fusion 360, reducing product development cycle time by 20%.</li><li>- Created manufacturable designs using Fusion 360 and validated them for Design for Manufacturing and Design for Assembly techniques.</li><li>- Reduced troubleshooting time by 35% by creating bills of materials and validation documents for inspections.</li></ul> |
| 07/2021 – 02/2022 | <p>MAS Intimates (Pvt) Ltd</p> <p><b>In-plant Trainee – Automation</b></p> <ul style="list-style-type: none"><li>- Created SolidWorks designs using Design for Manufacturing and Design for Assembly techniques.</li><li>- Designed and developed an attachment for sewing machines that reduced the time taken to cut the finished piece by 65%.</li><li>- Developed control systems for three ongoing projects to evaluate each conceptual method, using microcontrollers (Arduino, ATTiny84).</li></ul>                                        |

### Education

|                   |                                                                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04/2023 - Present | <p>Fachhochschule Südwestfalen, Soest</p> <p><b>- MSc. in Systems Engineering and Engineering Management (GPA – 1.9)</b></p> <ul style="list-style-type: none"><li>- Thesis: A Collaboration Workflow for Merging System Architecture Models - An Example from Agriculture Industry (Writing)</li></ul> |
| 2022              | <p>General Sir John Kotelawala Defence University, Sri Lanka</p> <p><b>- BSc. (Hons.) in Mechatronics Engineering (GPA – 1.8)</b></p> <ul style="list-style-type: none"><li>- Thesis: Development of Elbow Prosthetic Control System</li></ul>                                                          |

---

## Projects

|                                           |                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chess Game                                | <ul style="list-style-type: none"><li>- Developed a fully functional chess game with time management, move validation, move history and move notation features.</li><li>- Developed a reinforcement learning algorithm using DQN to train a chess agent.</li></ul>                                                                                 |
| IEEE BibTex Generator                     | <ul style="list-style-type: none"><li>- Developed a Python package to simplify the creation of BibTex entries for sources used in Latex.</li><li>- The package can automatically generate the IEEE BibTex entry for a given DOI.</li></ul>                                                                                                         |
| Remote Temperature Detection and Transmit | <ul style="list-style-type: none"><li>- Developed an application to detect temperature at a remote location and send using a GSM module.</li><li>- The application was developed for STM32 platform using C.</li><li>- The application has the capability to store the temperature readings and transmits using the GSM module via MQTT.</li></ul> |
| Grid Solving with Reinforcement Learning  | <ul style="list-style-type: none"><li>- Developed a custom grid environment and analysing performance of Q-learning, Deep Q-learning and A2C algorithms using Pytorch, OpenAI Gym and StableBaseline3.</li></ul>                                                                                                                                   |
| Prosthetic Elbow Concept                  | <ul style="list-style-type: none"><li>- Engineered a C++-based control system for precise control of an elbow joint using a brushless DC motor within a prosthetic arm.</li><li>- Designed and optimized the intricate gear system for the joint utilizing SolidWorks, ensuring seamless integration and functionality.</li></ul>                  |

---

## Technical Skills

|                       |                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------|
| Programming Languages | Python, C/C++, Arduino, MATLAB                                                         |
| Software              | Enterprise Architect, SolidWorks, Fusion 360, MATLAB Simulink, Autodesk Eagle, CubeIDE |
| Developer Tools       | Git, Azure DevOps, Docker, Jenkins, Flask                                              |
| Libraries             | Pytorch, Tensorflow, StableBaseline, OpenAI Gym                                        |

---

## Other Skills

|                 |                                                   |
|-----------------|---------------------------------------------------|
| IT              | Microsoft Office, Adobe Photoshop, HTML & CSS     |
| Hands-on-Skills | Handling Workshop tools, Soldering, PCB Designing |

---

## Languages

|         |          |
|---------|----------|
| German  | Beginner |
| English | Fluent   |

---

## Hobbies

- Playing Guitar
- Playing Computer Games